validity (for Newton's laws, the domain of speeds small compared to the speed of light). Physicists then struggle, experimentally and theoretically, to understand what is going on at the boundary of that domain of validity, and they finally formulate a new set of laws which is highly successful inside, near, and beyond the boundary (in Newton's case, *Einstein's special relativity*, valid at speeds approaching light as well as at low speeds). Then the process repeats. We shall meet the repetition in coming chapters: The failure of special relativity when gravity becomes important, and its replacement by a new set of laws called *general relativity* (Chapter 2); the failure of general relativity near the singularity inside a black hole, and its replacement by a new set of laws called *quantum gravity* (Chapter 13).

There has been an amazing feature of each transition from an old set of laws to a new one: In each case, physicists (if they were sufficiently clever) did not need any experimental guidance to tell them where the old set would begin to break down, that is, to tell them the boundary of its domain of validity. We have seen this already for Newtonian physics: Maxwell's laws of electrodynamics did not mesh nicely with the absolute space of Newtonian physics. At rest in absolute space (in the frame of the aether), Maxwell's laws were simple and beautiful—for example, magnetic field lines have no ends. In moving frames, they became complicated and ugly—magnetic field lines sometimes have ends. However, the complications had negligible influence on the outcome of experiments when the frames moved, relative to absolute space, at speeds small compared to light; then almost all field lines are endless. Only at speeds approaching light were the ugly complications predicted to have a big enough influence to be measured easily: lots of ends. Thus, it was reasonable to suspect, even without the Michelson–Morley experiment, that the domain of validity of Newtonian physics might be speeds small compared to light, and that the Newtonian laws might break down at speeds approaching light.

In Chapter 2 we shall see, similarly, how special relativity predicts its own failure in the presence of gravity; and in Chapter 13, how general relativity predicts its own failure near a singularity.

When contemplating the above sequence of sets of laws (Newtonian physics, special relativity, general relativity, quantum gravity)—and a similar sequence of laws governing the structure of matter and elementary particles—most physicists are driven to believe that these sequences are converging toward a set of ultimate laws that truly governs the Universe,